

Closing the Digital Support Gap

Evaluating a Community Technology Pilot Through Public Data

Google Data Analytics Professional Certificate — Capstone Project

Aaron Spoons · September 2026

Should a community organization fund a limited technology-support pilot?

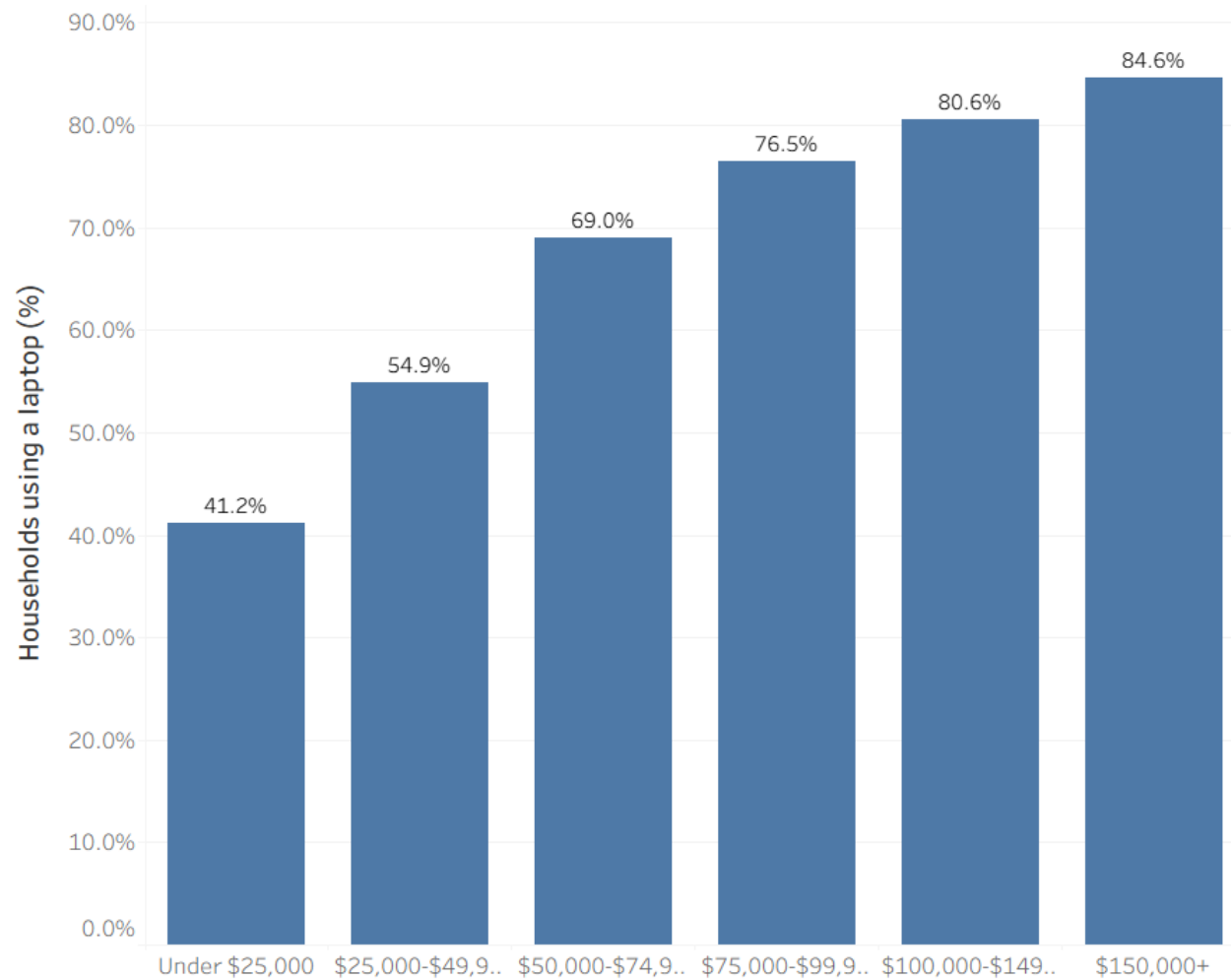
The proposed pilot combines:

- Device refurbishment and reuse
- Hands-on technical training
- Personalized digital support

Priority population: Older adults and other groups showing lower levels of digital engagement.

Decision standard: Does the available evidence show enough measurable need, repair potential, and public benefit to justify a limited pilot?

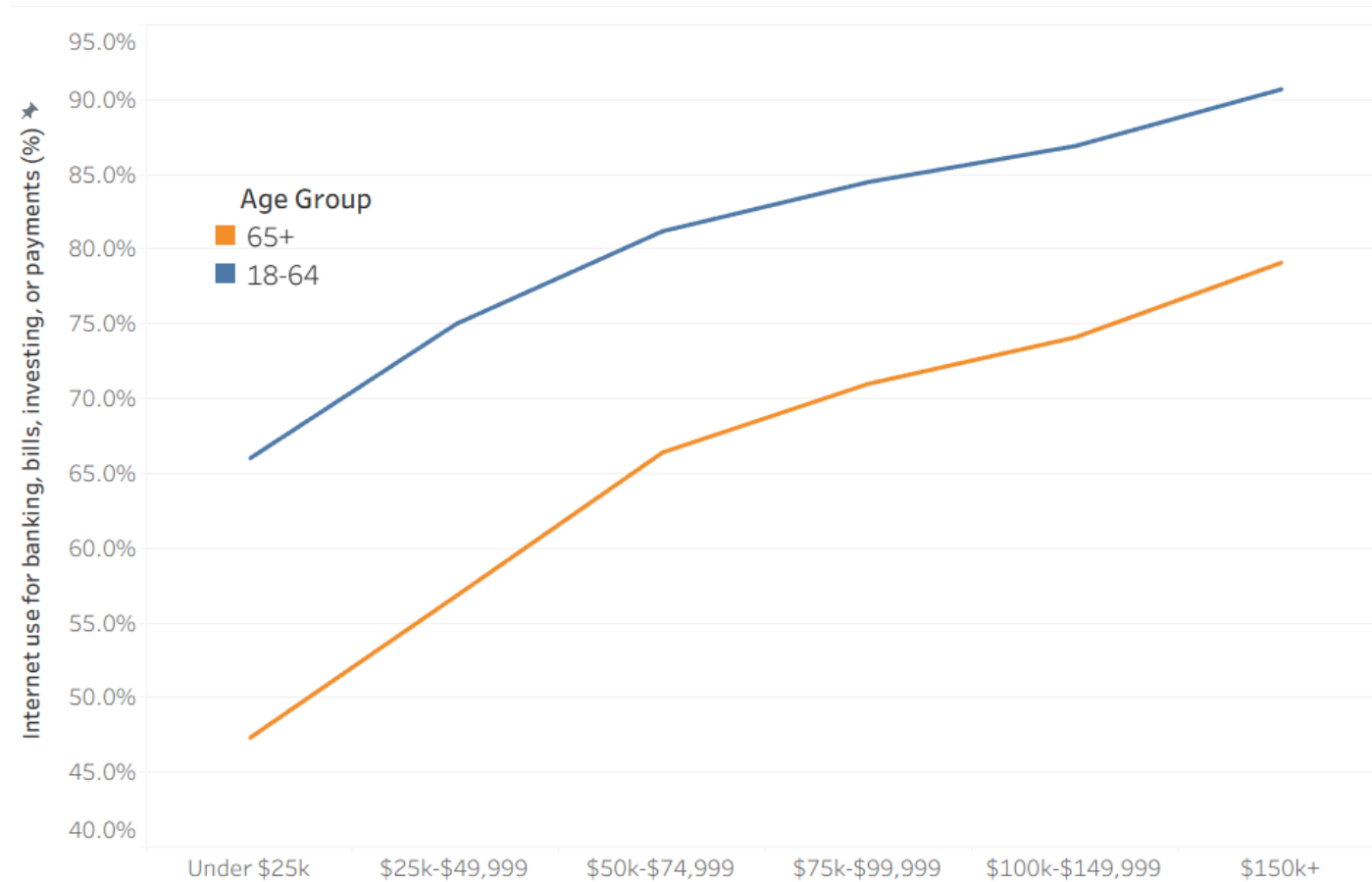
Laptop use rises sharply with household income



43.4 percentage-point gap

Laptop use rises from **41.2%** among households under \$25,000 to **84.6%** among households earning \$150,000 or more.

Older adults report lower use of internet-based financial services across income levels



63.2% vs 81.6%

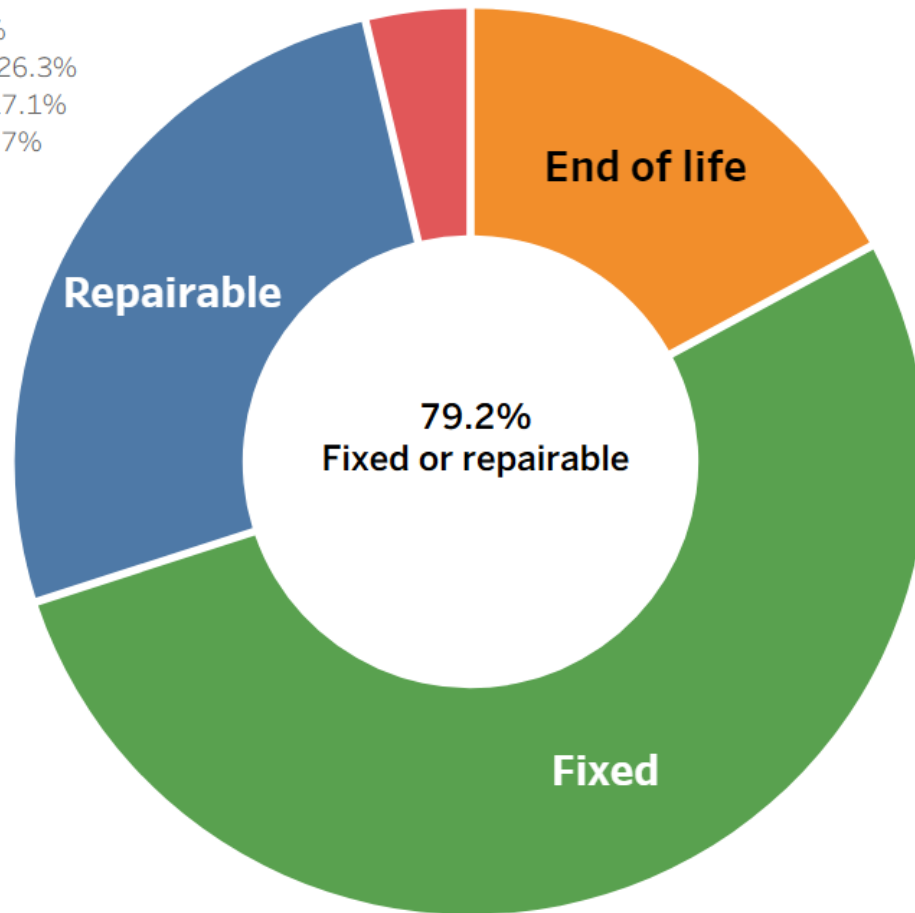
Internet financial-services use was 63.2% among adults age 65+, compared with 81.6% among adults ages 18–64.

The age gap persists across every household-income group shown.

Most core-device records were fixed or remained repairable

Core-device repair outcomes

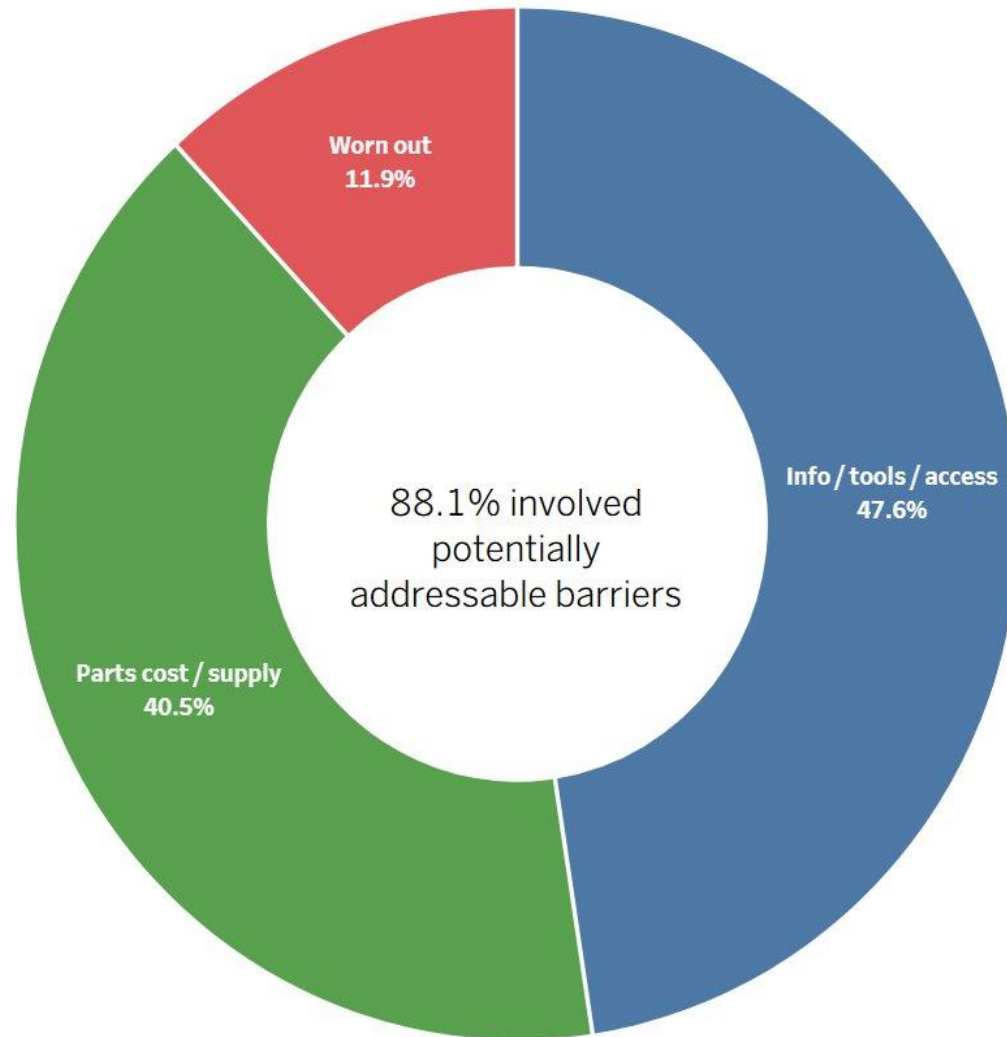
- Fixed — 52.9%
- Repairable — 26.3%
- End of life — 17.1%
- Unknown — 3.7%



52.9% of records were fixed and another **26.3%** remained repairable.

*“Repairable” does **not** mean the device was successfully repaired during the recorded attempt.*

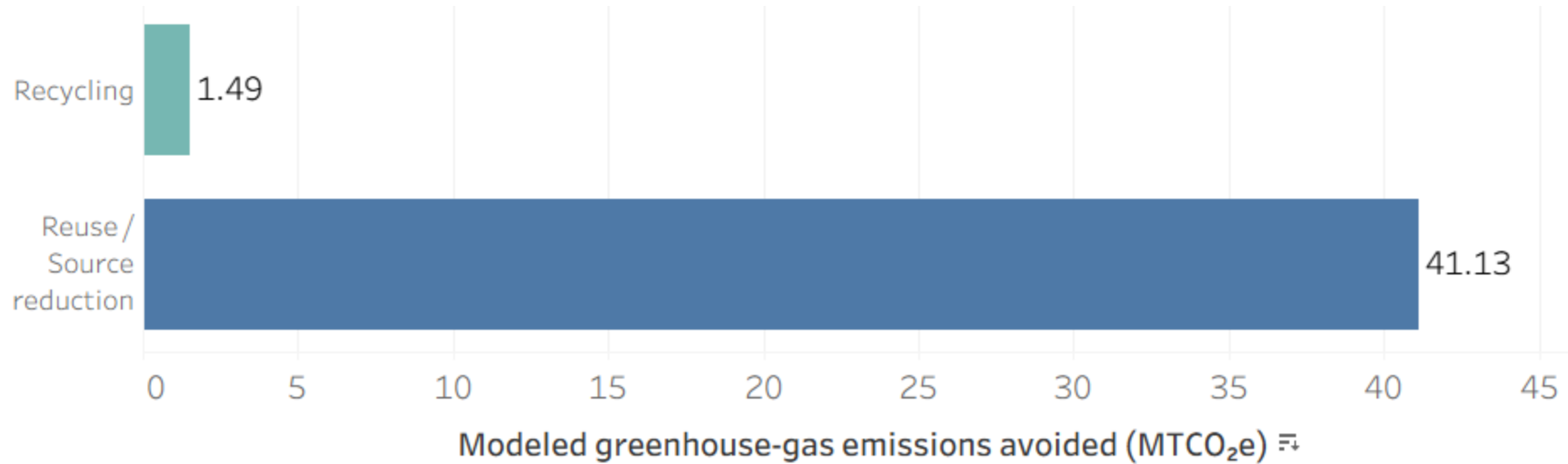
Most documented end-of-life barriers were potentially addressable



Most documented barriers involved **information, tools, product access, parts cost, or parts availability** rather than simple device wear-out.

*Barrier information was documented for only **29.0%** of core-device end-of-life records.*

Reuse offers a much larger modeled emissions benefit than recycling



41.13 MTCO₂e

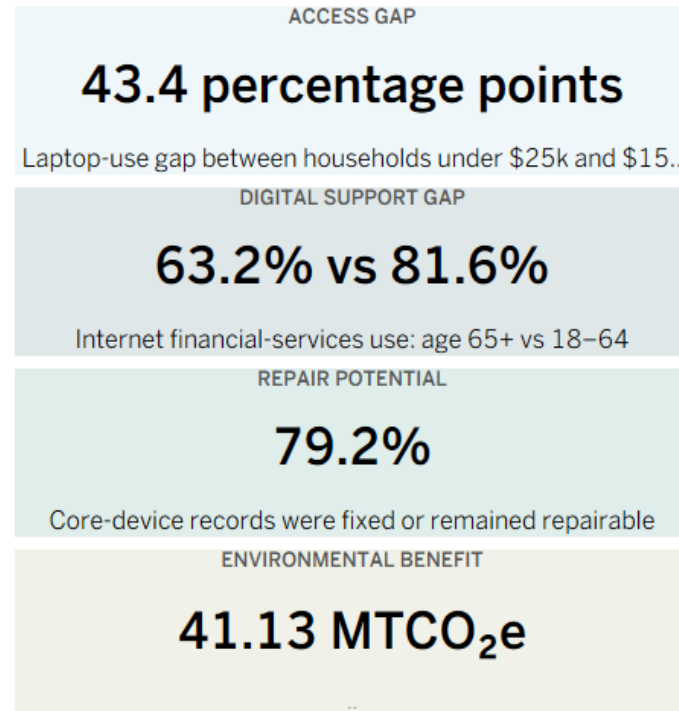
Reusing **500 portable devices** instead of landfilling them produced a modeled benefit of **41.13 metric tons of carbon dioxide equivalent (MTCO₂e)**.

Recycling the same modeled device mass avoided **1.49 MTCO₂e**.

*These are **scenario estimates**, not measured program outcomes.*

Source: U.S. Environmental Protection Agency (EPA), Waste Reduction Model (WARM), electronics factors; author's scenario analysis. Assumes 500 portable electronic devices averaging 2.5 kg each. *Results are modeled estimates, not observed program outcomes.*

Integrated Evidence for a Community Technology Pilot



Recommendation

The evidence supports funding a limited, measurement-focused pilot combining device refurbishment and reuse, hands-on technical training, and personalized digital support.

The pilot should be evaluated before considering broader implementation.

Beyond This Pilot

Fraud-awareness support

Fraud-awareness support warrants its own pilot; evidence suggests intervention format (one-time vs. repeated/trusted contact) matters more than whether education happens at all, which this study didn't test.

Environmental impact

Environmental impact warrants better modeling, not a separate pilot; real WARM estimates should be calculated from the refurbishment pilot's actual device data once it exists, replacing the illustrative 500-device scenario.

Methodology, Limitations & AI-Assisted Workflow

METHOD

- National Telecommunications and Information Administration (NTIA) 2023 survey data analyzed using survey weights
- Open Repair records filtered to laptops, desktops, mobile phones, and tablets
- SQL used to independently validate key Python-derived results
- U.S. Environmental Protection Agency (EPA) Waste Reduction Model (WARM) used for a defined 500-device reuse scenario
- External evidence manually reviewed before inclusion

LIMITATIONS

- NTIA measures reported technology and internet use, not digital proficiency
- Open Repair data are self-selected and not nationally representative
- Repair-barrier information was available for only 29.0% of core-device end-of-life records
- WARM results are modeled estimates, not observed program outcomes
- Findings support a pilot test, not immediate full-scale implementation

AI USE

- Artificial intelligence tools assisted with source discovery, code drafting, workflow planning, summarization, and documentation.
- External sources and analytical outputs were manually reviewed and validated.
- Final decisions regarding data selection, methodology, interpretation, limitations, and recommendations were made by the analyst.